

Ritika Kalia

Computer Science & Engineering | Chandigarh College of Engineering & Technology

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EDUCATION

B.E. in Computer Science & Engineering, Chandigarh College of Engineering & Technology **2024–2027**
CGPA: 9.45 / 10.0 (through 6th semester)

Diploma in Computer Science & Engineering, Chandigarh College of Engineering & Technology **2022–2024**
Percentage: 86%

TECHNICAL SKILLS

Programming: Python, C, SQL, JavaScript

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Web Development: Django, Flask, HTML, CSS, Bootstrap, AJAX

Machine Learning / Deep Learning: PyTorch, Hugging Face Transformers

Tools: Git, GitHub, VS Code, Google Colab, Kaggle, Jupyter Notebook

PROJECTS

Full-Stack E-Commerce Platform (StellarCart)

Python, Django, Bootstrap, HTML, CSS, JavaScript | GitHub

- Built a full-stack Django application with Google authentication, product catalog, cart, checkout, and order management.
- Designed related database models using Django ORM and implemented AJAX-based cart updates with CSRF protection.
- Integrated Razorpay test-mode payments with server-side amount calculation and payment-signature verification, alongside Cash on Delivery.

Multilingual Legal QA with Cross-LLM Refinement

Python, PyTorch, Qwen2.5-3B, QLoRA, Flask | GitHub

- Fine-tuned Qwen2.5-3B with QLoRA into four language adapters (English, Hindi, Punjabi, Nepali) for Indian legal Q&A.
- Built a retrieval step using BGE embeddings for English and LaBSE for other languages to fetch relevant context per question.
- Built a two-model pipeline where Qwen2.5-7B critiques and refines the adapter model's draft, selecting the better answer using similarity to a reference answer.
- Evaluated outputs with multiple metrics; built a Flask web app for inference and a batch-evaluation script.

MMVDNet: Video Violence Detection System using VideoMAE

Python, PyTorch, VideoMAE, Flask | GitHub

- Built a fight-scene detection system using a Kinetics-pretrained VideoMAE backbone with multi-clip sampling on the RWF-2000 benchmark.
- Built a Flask web app supporting short-video uploads and YouTube Shorts links, with real-time violence detection.

PUBLICATIONS & ACHIEVEMENTS

- IEEE ICIR 2026 – Research Publication & e-Poster Presentation:** “Real-Time Privacy Guardianship in XR Metaverses via AI-Driven Behavioral Analytics”; developed BehaviorShield, a real-time behavioral analytics system for XR privacy intrusion detection, achieving a **mean ROC-AUC of 0.967 ± 0.009** under player-level Leave-One-Subject-Out (LOSO) cross-validation. [IEEE Xplore] [GitHub]
- 3rd Position, Business Blueprint 2025 (IIC), CCET:** Pitched “Women Safety Analytics System,” a multi-modal AI/ML surveillance solution.

EXPERIENCE

CS Soft Solutions (I) Pvt. Ltd.

Jul 2023 – Aug 2023

Python Programming — Industrial Training: Completed industrial training covering Python programming fundamentals

LEADERSHIP & ACTIVITIES

- Chairperson**, ACM-W CCET Student Chapter Jul 2025 – Jul 2026
- Technical Team Member**, CCET APRATIM 2025 (College Tech Fest) Sep 2025 – Nov 2025
- Design Team Member**, CCET College Website Oct 2024 – Aug 2025
 - Designed departmental web pages using Figma as part of the college website team.